BeachWell

Our CECS 458/558 term project, Fall 2026. Team: Vo Hong Vy Le and Deborah Shaw

BeachWell gives **every week of your semester a workload score** from 0 to 100. It reads a student's Canvas deadlines, warns which weeks will be an overload, builds a study plan that spreads the work out, and suggests campus events to recharge during midterms and finals.

The problem

- **School is the main source of stress.** In a national survey, 87% of college students named education as their primary stressor.
- **Crunch weeks sneak up.** Canvas lists due dates, but not how much work a week really holds. Students notice only when three big deadlines land in four days.
- **Help exists but gets missed.** CSULB (42,000+ students, many commuters) already runs free de-stress events and CAPS workshops, but nothing tells a stressed student which one fits their free hour.

How it works

1 Connect

Paste the Canvas Calendar Feed link and class times. Optional 20-second daily check-in (sleep, stress).

2 Score

Each week gets a score from 0 to 100 and a level (Light, Moderate, Heavy, Overload), with reasons ("3 big deadlines in 4 days").

3 Plan

Big assignments become study blocks moved into lighter weeks, with breaks and small habit tips. Exports to Google or Apple Calendar.

4 Recharge

For each heavy week, 2–3 campus events that match the student's interests and fit their free time.

What makes it special

- **What-if slider.** Drag "start 10 days earlier" and watch an overload week's score drop, live. Our best demo moment.
- **Learns your pace.** Log how long work really took, and estimates adjust: "you take 30% longer on essays."
- **Campus Stress Radar.** Anonymous totals show ASI, CAPS and professors when whole groups head into an overload week, and which deadlines collide. Groups under 10 students are never shown.



The workload screen in our working prototype (demo student "Vy," week 8).

How it reaches students

BeachWell is free for students and would be offered as a campus service through Student Affairs and ASI, starting with a small pilot. Running costs are low (AI usage and web hosting), and it needs no new campus systems because it uses data students already have in Canvas.

> **Safety rule:** BeachWell plans and recommends; it never diagnoses. If a check-in note shows signs of crisis, the app immediately shows the 988 Lifeline, CSULB CAPS and 911. That message is fixed text, never written by AI.

Where the AI is

Part	What it does
Deadline Agent	Reads messy Canvas titles ("Lab 4," "Milestone 2") and estimates the type and hours of work. Gemini when a key is set; rules otherwise.
Workload score	Turns hours, deadline clusters, sleep and stress into a weekly score from 0 to 100 with reasons.
Pace model	Learns each student's speed per assignment type from logged hours.
Planner Agent	Places study blocks: never during class, never after a deadline, max 90 minutes, breaks included.
Event Recommender	Matches events to interests by meaning (embeddings), checks free time, and writes a one-line reason.
Safety Agent	Screens every free-text note for signs of crisis.
Campus Radar	Combines opted-in workload scores into anonymous heatmaps and alerts.

First results (compared with a simple non-AI method)

What we measured	Simple method	BeachWell
Error in hour estimates (60 assignments)	2.6 h	**1.75 h**
Finding the real crunch week (40 test semesters, F1 score)	0.67	**1.00**
Estimate error late in term, after learning pace	1.4 h	**0.77 h**
Hours in the busiest week, with the study plan	41.5 h	**30.1 h**
Overload weeks per student, with the study plan	3.2	**1.6**
Event picks that match the student, top 3 (24 test students)	0.46	**0.47**

These numbers use practice labels and made-up test semesters, and they come from offline mode (no Gemini). Before the final report we re-label the data ourselves and re-run with Gemini.

Where we are

Done Working prototype: all 7 parts, web app, 32 tests

Done Evaluation script and first results

To do Get a Gemini key and re-run the evaluation

To do Both of us re-label the test data

To do Put the code on GitHub (everyone commits their own part)

To do Replace sample event dates with real ones

Deadlines

Fri Oct 16	Phase 1 report (max 4 pages)
Fri Nov 20	Phase 2: prototype, GitHub docs, 2-page report
Fri Dec 11	Phase 3: 10-minute pitch video and final report

Who does what (proposed)

Vo Hong Vy Le	Backend, AI agents, evaluation
Deborah Shaw	Web app, market research, event list, reports
Both	Label the test data separately, so we can measure how well we agree

Try it yourself (5 minutes)

Needs Python 3.10+. Unzip `beachwell.zip`, then:

```
cd backend
pip install -r requirements.txt
python scripts/seed.py --reset
uvicorn beachwell.app:app --port 8000
```

Open **localhost:8000** and click **Open the demo**. No API key needed. Full details are in `README.md`.